

Why `cellCentredReconstruct` states fail on a propagating TNNP front while `gaussPointODE` works

A diagnosis of ignition and conduction failure in
`highOrderElectroActivationFoamImplicitPETSc`

Solver diagnosis note

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Abstract

The solver offers two ways of handling the ionic state variables $\mathbf{s} = (s_1, \dots, s_N)$ in the high-order ionic-current quadrature: `gaussPointODE` (the legacy path, one stiff ODE integration per Gauss point) and `cellCentredReconstruct` (one ODE per cell centre, states then reconstructed at the Gauss points by a high-order LRE of order `p1/p2/p3`). On the Niederer *et al.* (2012) benchmark with the TNNP cell model, `cellCentredReconstruct` fails to produce a propagating activation as the time step is refined ($\Delta t = 10^{-5}$ s): the wavefront neither ignites (with the nominal stimulus) nor conducts (even with a stronger stimulus), whereas `gaussPointODE` ignites and propagates on the same mesh and Δt . This note shows, with three controlled discrimination runs and a short mathematical argument, that this is *not* a coding bug, nor a time-scheme (Crank–Nicolson) artefact, nor a Δt -scaling error. It is an intrinsic accuracy limitation of evaluating a strongly non-linear ionic kinetics at the *cell-average* membrane potential: by Jensen’s inequality the cell-average sodium activation under-estimates the pointwise average, weakening the regenerative I_{Na} current at the sharp upstroke. Raising the reconstruction order (`p2`, `p3`) does not help, because the order only affects how the already-computed states are interpolated, not the potential that drove the ODE.

1 Governing equations and the two state-integration modes

The monodomain problem solved for the transmembrane potential V_m is

$$\chi C_m \frac{\partial V_m}{\partial t} = \nabla \cdot (\mathbf{D} \nabla V_m) - I_{\text{ion}}(V_m, \mathbf{s}) + I_{\text{ext}}, \quad \frac{\partial \mathbf{s}}{\partial t} = \mathbf{g}(V_m, \mathbf{s}), \quad (1)$$

where $\mathbf{s}$ collects the N gating variables and ion concentrations of the ten Tusscher–Noble–Noble–Panfilov (TNNP) cell model, $\mathbf{g}$ is its stiff right-hand side, and $I_{\text{ion}} = \phi(V_m, \mathbf{s})$ is the (algebraic) total ionic current. The reaction source of (1) is integrated over each cell Ω_c with a high-order quadrature at points $\{x_q\}$ and weights $\{w_q\}$ ($\sum_q w_q = |\Omega_c|$),

$$\overline{I_{\text{ion}, c}} = \frac{1}{|\Omega_c|} \int_{\Omega_c} I_{\text{ion}} \, d\Omega \approx \frac{1}{|\Omega_c|} \sum_q w_q \phi(V_m(x_q), \mathbf{s}(x_q)). \quad (2)$$

The membrane potential at the Gauss points, $V_m(x_q)$, is obtained in both modes by the same high-order reconstruction of the cell-centred field. The two modes differ *only* in how the state vector $\mathbf{s}(x_q)$ that enters (2) is obtained.

`gaussPointODE` (legacy). The stiff ODE is integrated *independently at every Gauss point*, driven by the locally reconstructed potential:

$$\mathbf{s}_q^{n+1} = \text{ODEsolve}(\mathbf{s}_q^n, V_m(x_q); \Delta t), \quad \mathbf{s}(x_q) \equiv \mathbf{s}_q^{n+1}. \quad (3)$$

Cost: N_q stiff integrations per cell.

cellCentredReconstruct (new). The ODE is integrated *once per cell*, driven by the cell-centre potential $\bar{V}_{m,c} \equiv V_m(x_c)$, and the resulting cell-centred states are then reconstructed to the Gauss points by an LRE of order p :

$$\mathbf{s}_c^{n+1} = \text{ODEsolve}(\mathbf{s}_c^n, \bar{V}_{m,c}; \Delta t), \quad \mathbf{s}(x_q) \equiv \hat{\mathbf{s}}^{(p)}(x_q) = \mathcal{R}^{(p)}[\{\mathbf{s}_{c'}^{n+1}\}_{c' \in \text{stencil}}](x_q). \quad (4)$$

Cost: one stiff integration per cell (the motivation: I_{Na} -limited TNNP integrations dominate the run time, so 1 vs N_q ODEs per cell is a large saving).

The essential contrast is already visible in (3)–(4): in `gaussPointODE` the non-linear map $V_m \mapsto \mathbf{s}$ is applied to the *pointwise* $V_m(x_q)$; in `cellCentredReconstruct` it is applied to the *cell-average* $\bar{V}_{m,c}$, and only the (already-computed) result is interpolated.

2 Observed symptom

Benchmark: 2-D slab, structured hexahedra, $\Delta x = 0.5$ mm (40×16), TNNP, a 1.5 mm corner stimulus box of intensity 50 000 A/m³ for 2 ms, activation threshold 0 mV, Crank–Nicolson, Picard coupling, RKF45 for the states. With `cellCentredReconstruct` at order p1 (“CCp1”):

- at $\Delta t = 10^{-4}$ s the activation ignites and propagates (diagonal points activate in sequence);
- at $\Delta t = 10^{-5}$ s *nothing* activates — every diagonal sample, including the stimulus site itself, stays at $t_{\text{act}} = 0$.

A finer (more accurate) time step producing *no* activation is the counter-intuitive signature that must be explained.

3 Discrimination experiments

Three controlled runs isolate the cause. All use the same mesh (40×16 , $\Delta x = 0.5$ mm), $\Delta t = 10^{-5}$ s, endtime 10 ms (long enough: the stimulus site fires at ~ 1.3 ms when it triggers). “Ignition” means the stimulus site (diagonal index 0) crosses the threshold; “conduction” means the front advances along the diagonal.

Run	Configuration	Ignition	Conduction	Evidence
A	<code>gaussPointODE</code> (states=N0), CN	yes	yes	pt. 0 @ 1.20 ms; pts. 0–4 active, front ~ 4.5 mm
—	<code>cellCentredReconstruct</code> CCp1, CN (original)	no	—	all $t_{\text{act}} = 0$; max $V_m = -27$ mV
B	<code>cellCentredReconstruct</code> CCp1, backwardEuler	no	—	all $t_{\text{act}} = 0$; max $V_m = -27$ mV
C	<code>cellCentredReconstruct</code> CCp1, CN, stimulus $\times 4$	yes	no	pt. 0 @ 0.60 ms; pts. 0–2 only, front stalls ~ 2.3 mm

Table 1: Discrimination runs at $\Delta t = 10^{-5}$ s. `gaussPointODE` ignites and conducts; `cellCentredReconstruct` does neither with the nominal stimulus, and even a $4\times$ stimulus only ignites without sustaining conduction.

The three runs rule out the “obvious” suspects and localise the cause:

Not the time scheme.

Run B replaces Crank–Nicolson by the fully dissipative backward Euler and still fails identically (max $V_m = -27$ mV). Crank–Nicolson is exonerated.

Not a Δt -scaling bug.

The θ -scheme is Δt -consistent ($A = M/\Delta t - \theta L$, $B = M/\Delta t + (1 - \theta)L$, source weighted by θ , $(1 - \theta)$ with no spurious Δt factor), and the stimulus is applied over a fixed physical window $[t_0, t_0 + 2$ ms] independent of Δt . Both Δt inject the same charge.

It is the state mode.

Run A is identical to the failing case except that the states are solved at the Gauss points (`gaussPointODE`). It ignites at 1.20 ms and conducts. The only changed ingredient is (3) vs (4).

4 Root cause: averaging a non-linear kinetics

4.1 The sodium activation is convex at the foot of the upstroke

The fast depolarisation is carried by the sodium current $I_{\text{Na}} = g_{\text{Na}} m^3 h j (V_m - E_{\text{Na}})$, whose activation gate relaxes towards the steep sigmoid

$$m_\infty(V_m) = \left[1 + \exp((V_{1/2} - V_m)/k) \right]^{-1}, \quad V_{1/2} \approx -40 \text{ mV}, \quad k \approx 6 \text{ mV}. \quad (5)$$

Below its inflection ($V_m < V_{1/2}$) — exactly the sub-threshold “foot” where ignition is decided — m_∞ is *convex*. Over a cell Ω_c that straddles the wavefront, $V_m(x)$ ranges from resting on the trailing side to strongly depolarised on the leading side.

4.2 Jensen’s inequality: `cellCentredReconstruct` under-shoots the reactive current

Let $\langle \cdot \rangle = \frac{1}{|\Omega_c|} \sum_q w_q(\cdot)$ denote the cell-quadrature average, so $\bar{V}_{m,c} = \langle V_m \rangle$. `gaussPointODE` evaluates the kinetics pointwise and averages the *outcome*; `cellCentredReconstruct` averages the *input* first. For the convex m_∞ ,

$$\underbrace{\langle m_\infty(V_m) \rangle}_{\text{gaussPointODE (effective)}} \geq \underbrace{m_\infty(\langle V_m \rangle)}_{\text{cellCentredReconstruct}} = m_\infty(\bar{V}_{m,c}). \quad (6)$$

Because $I_{\text{Na}} \propto m^3$, the gap in (6) is amplified cubically. `cellCentredReconstruct` therefore systematically *under-estimates* the open-sodium fraction, hence the inward (depolarising) current, precisely where V_m varies most within a cell — at the stimulus edge and along the upstroke. In `gaussPointODE` the Gauss points sitting on the depolarised side of the cell each run their own ODE, see a high local $V_m(x_q)$, open their sodium channels, and contribute a strong inward current to (2); those sub-cell “hot spots” are exactly what `cellCentredReconstruct` averages away before the kinetics ever sees them.

4.3 Two failure modes

1. **Ignition failure.** With the nominal stimulus the site depolarises to only $V_m \approx -27 \text{ mV}$ (Runs orig/B) and relaxes back: the weakened I_{Na} never becomes regenerative, so the all-or-none upstroke does not fire. A $4\times$ stimulus (Run C) forces V_m past threshold at the site (0.60 ms), confirming the nominal stimulus was rendered *effectively sub-threshold* by the averaging.
2. **Conduction block.** Even once ignited (Run C), the front stalls after 2–3 cells. Propagation requires each activating cell to supply enough inward charge (*source*) to bring the downstream cells (*sink*) to threshold — the cardiac “safety factor”. By (6), `cellCentredReconstruct` lowers the source at the very front where it is needed, driving the safety factor below unity: a numerically induced source–sink mismatch, i.e. conduction block.

5 Why the coarse time step “worked”

At $\Delta t = 10^{-4} \text{ s}$ the implicit step advances the depolarisation in one large jump: the reaction/stimulus source acts over a big $\Delta t/M$ increment, producing a transient over-shoot of $\bar{V}_{m,c}$ that can exceed threshold even with the under-estimated I_{Na} . As $\Delta t \rightarrow 0$ the computed trajectory converges to the *true* (accurate) one, which under `cellCentredReconstruct`’s weakened current stays sub-threshold — so the deficiency, masked by coarse-step over-shoot, is revealed. Consequently the $\Delta t = 10^{-4} \text{ s}$ CCp1 activation is an artefact of temporal under-resolution and its activation times should not be trusted; it is worth checking that they agree with `gaussPointODE` at the same Δt .

6 Why higher reconstruction order does not help

Modes p2 and p3 change only the operator $\mathcal{R}^{(p)}$ in (4) — how the cell-centred states $\mathbf{s}_c^{n+1}$ are interpolated to the Gauss points. They do *not* change the fact that $\mathbf{s}_c^{n+1}$ was produced by integrating the ODE with the cell-average input $\bar{V}_{m,c}$. The bias in (6) is committed *inside* the ODE solve, before any reconstruction; a higher-order interpolation merely refines an already-biased quantity. Empirically and by construction, CCp2/CCp3 inherit the same ignition and conduction failure as CCp1 on this front.

7 When `cellCentredReconstruct` is valid, and recommendation

(6) is an equality when V_m is (nearly) linear across the cell, i.e. when the sub-cell variation is small compared with the curvature scale of m_∞ . This is the condition

$$h_{\text{cell}} \ll w_{\text{front}}, \quad (7)$$

the cell size being much smaller than the upstroke width. The cardiac upstroke is $\sim 0.5\text{--}1$ mm wide; at $h = 0.5$ mm a cell spans the whole front and `cellCentredReconstruct` fails, whereas $h \lesssim 0.1$ mm would resolve it — at a cell count that erodes the very cost advantage `cellCentredReconstruct` was meant to provide.

- **Propagation with a stiff cell model (Niederer/TNNP):** use `gaussPointODE` (`stateIntegrationMode gaussPointODE`, or `states="NO"` in the triad sweep). It resolves the sub-cell kinetics and both ignites and conducts.
- **Smooth reaction fields** (e.g. the manufactured MMS case of the FDA solver, where V_m varies little within a cell): `cellCentredReconstruct` is accurate and delivers its intended $N_q \rightarrow 1$ ODE speed-up.
- **Possible hybrid (future work):** integrate the ODE at the Gauss points only where $|\nabla V_m|$ is large (the moving front) and at the cell centre with reconstruction elsewhere (resting/plateau). This would recover most of the speed-up while preserving accuracy at the front.

Bottom line. `cellCentredReconstruct` is not broken and its implementation is correct; it simply trades sub-cell resolution of the ionic kinetics for speed. That trade is safe for smooth reactions but invalid for the steep, self-sustaining upstroke of a real cardiac action potential, where averaging the membrane potential before the sodium kinetics suppresses ignition and blocks conduction. For the Niederer benchmark, `gaussPointODE` is the appropriate mode.